Designed by:

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The Business Model Canvas

Key Partnerships

- Educational institutions and STEM schools providing laboratory environments
- University laboratories supporting benchmarking and validation
- Recycling and e-waste sources supplying reusable electronic components
- Academic mentors and capstone supervisors offering technical oversight
- Public laboratories and research centers focused on water analysis

Key Activities

- Automated system
- Closed loop system
- Automated feeding system
- Providing plant nutrients
- saving the lives of living organisms

Key Resources

- Recycled materials
- Electric components
- Structural frame and containment materials
- Electrical control and power-regulation systems
- living organisms

Value Propositions

- Automated closed-loop system
- providing nutrients for Tilapia from the fish wastes
- Filtrating the fish wastes
- High accuracy results, due to sensor readings and actuator actions
- Save the living organisms' conditions by dealing with their setpoints
- Supports national goals related to water pollution and resource efficiency
- Keeping the money fund
- Save the water resource from overuse

Customer Relationships

- Direct engagement with educational and laboratory users
- Instructor-led implementation and guided usage

Channels

- STEM high school for boys, 6th of October
- STEM exhibitions, science fairs, and academic showcases
- Educational programs focused on environmental analysis

Customer Segments

- Secondary schools and STEM education programs
- Universities and academic research laboratories
- Environmental monitoring and water-quality testing facilities
- Institutions in resource-limited or developing regions
- Students and researchers conducting analytical chemistry experiments

Cost Structure

- Recycled materials of zero cost
- Assembly and fabrication expenses
- Testing, calibration, and validation costs
- Maintenance and component replacement over time
- Electric components cost

Revenue Streams

- Sale of low-cost Aqua Green laboratory units
- Distribution to educational institutions as teaching equipment
- Institutional adoption for routine water-quality testing
- Potential future upgrades with enhancedcondition control systems